

Remembering Autobiographically Consequential Experiences: Content Analysis of Psychologists' Accounts of Their Lives

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The autobiographies of 49 eminent psychologists were content analyzed in terms of autobiographically consequential experiences (ACEs). Most memories for ACE were not single episodes. Episodic ACEs did, however, share many characteristics of "flashbulb" and vivid memories elicited in studies using more traditional experimental procedures. Memories were concentrated during the college and early adult years. Thus, as in other autobiographical memory studies that have used older subjects, there was a pronounced reminiscence effect. These results were considered in light of Erikson's theory of adult personality development.

In the last decade, there has been a striking increase in the amount of research dealing with the recall of personally meaningful experiences occurring in natural settings. Although a few scholars (Banaji & Crowder, 1989) have argued that such studies cannot advance psychologists' understanding of memory process, most researchers have indicated that this increased interest is at least a useful complement to traditional laboratory approaches to memory (e.g., see Linton, 1982). Strong advocates for more naturalistic methods view the study of autobiographical memory as an overdue attempt to address important questions about memory that have been difficult or impossible to study in the laboratory (Neisser, 1978, 1988). These contrasting rationales for studying the memory of life events have produced a wealth of studies covering an extraordinary variety of experiences. To name a very few, there have been studies evaluating a person's memory for lyrics (Wallace & Rubin, 1988), television shows (Squire & Fox, 1980), a school-learned foreign language (Bahrick, 1984), the very earliest experiences (Fivush, Gray, & Fromhoff, 1987), public events (N. R. Brown, Rips, & Shevell, 1985), ordinary everyday happenings (Barclay & Wellman, 1986), and even real-life simulations and reenactments (Loftus, Manber, & Keating, 1983).

In spite of the broad range of experiences examined, there is very little information about memories of experiences that are viewed as central or pivotal in a person's life. It seems, intuitively, that an experience that is felt to have been of central importance in one's life must occupy a special position in that person's memory. Yet currently, researchers know little that goes beyond common sense or the anecdotal observations of the clinic. Quantitative investigations are rare.

A special role for *consequential* life experiences (experiences that are felt to have produced substantial and lasting effects) has occasionally been proposed. For example, Brown and Kulik (1977) implicated consequentiality as one of the crucial conditions for the formation of "flashbulb" memories (memories that are experienced as especially vivid—as if a flashbulb went off at the time of the experience and an internal photograph were taken of one's circumstances at the time). Although the idea that consequentiality should play a special role in the formation of a memory is intuitively appealing, there are reports of memorable experiences rated as having no more than a low or moderate level of consequentiality. For example, Rubin and Kozin (1984), who asked 58 undergraduates to describe their three clearest autobiographical memories, found that their subjects placed most of their experiences in the three bottommost positions of a 7-point scale of life consequentiality. Similarly, a detailed examination of the memories of 4 women for a 14-day period revealed that most of the remembered events "were not perceived as being loaded with personal significance" (Barclay & DeCooke, 1988, p. 101).

One way to understand these findings is to conclude that even though consequentiality may be a sufficient condition for memorability, it is not required. Experiences may be memorable for various reasons. There is abundant evidence (e.g., see Rubin & Kozin, 1984) that a strongly emotional experience may become memorable without having any long-term consequences. Still, the fact that consequentiality appears to be a sufficient condition for memorability, coupled with the fact that so little is known about its role or character, suggests a significant gap in the emerging framework of knowledge regarding autobiographical memory.

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Identifying Autobiographically Consequential Experiences

For reasons of convenience, investigators studying autobiographical memory have tended to use college students as sub-

jects (see Fitzgerald, 1988, for an exception), and for most memory studies this is a reasonable strategy. College students, however, may actually be ill suited for studying autobiographically consequential experiences (ACEs). We do not now know of any ways in which ACEs carry markers that inform us of their future consequentiality; they are identified best through the long lens of hindsight. Because of the advantage afforded by hindsight, younger subjects such as college students are handicapped in their ability to identify ACEs. Indeed, it is probably true that the suitability of a person to be a subject in a study of ACEs to a great extent depends upon his or her age—the older the better. Perhaps memories of events that turn out to be seen as consequential are different from those that do not, even immediately after the event. Such differences could be identified by comparing memory reports of events near the time of their experience on the basis of their later consequentiality, as assessed by the individual.

A reasonable first step, though, is to study subjects capable of articulating the consequentiality of events in their lives. Furthermore, these subjects should operate under a common set of instructions to direct them in their review of their life's course, directions that ask them to focus on the ACEs in their lives. Fortunately, we were able to identify a suitable, preexisting source of data in the autobiographies of persons whose lives are of great interest to us: distinguished psychologists who contributed to the three most recent volumes of *A History of Psychology in Autobiography* (Boring & Lindzey, 1967; Lindzey, 1974, 1979). Because this series included only two female psychologists in these three volumes, and because it seemed problematic to exclude the voices of women at this early stage in the study of memories of ACEs, we also examined the autobiographies of an additional 16 women, included in *Models of Achievement: Reflections of Eminent Women in Psychology* (O'Connell & Russo, 1983). Certainly, these autobiographies were prepared under special circumstances and with a particular focus. All autobiographies are, though, so the critical issue for us was to identify ones that were comparable, and these clearly are. Future research can directly assess the effects of particular audiences, instructions, and samples on the content of autobiographical documents.

Content analysis of existing personal documents is a technique increasingly used in personality research, but it is used much less widely in studying memory processes. These content-analysis techniques have been successfully applied to life history narratives and have generated a rich data source for studies of human personality (McAdams, 1985, 1988; Runyan, 1982). More generally, content-analysis techniques have enabled personality researchers to extract quantitative data from qualitative autobiographical and other narrative accounts (Krippendorff, 1980; Stewart, Franz, & Layton, 1988; Winter & Carlson, 1988). Other researchers studying autobiographical memory have relied on content-analysis techniques (e.g., Brown & Kulik, 1977; Pillemer, Rhinehart, & White, 1986). However, they have applied these techniques to memories assessed in highly structured laboratory settings. In this study, we extended the use of these techniques to memories reported in the course of narratives prepared for other purposes. In this way, we analyzed naturally occurring autobiographical memories in a nonreactive

way and tested the range of applicability of concepts proven useful in laboratory studies.

Issues in Autobiographical Memory

The first question, then, is to what extent do ACEs resemble the episodic, one-moment-in-time events that give rise to flashbulb (Brown & Kulik, 1977) or vivid (Rubin & Kozin, 1984) memories, the events that constitute the bulk of the autobiographical memory literature? Although intuition and some limited data (Pillemer et al., 1986) suggest that ACEs may extend over days, weeks, months, or even years (rather than occurring in fleeting episodes), this idea has not been fully investigated. And if ACEs are not episodic in nature, is this a difference that makes a difference? In what ways do episodic and nonepisodic ACE memories resemble and differ from each other?

To answer these questions, we need to compare episodic and nonepisodic ACE memories in terms of the characteristics commonly used to describe memories (Brown & Kulik, 1977; Pillemer, 1986; Rubin & Kozin, 1984). These characteristics include description of location, presence of affect, ongoing activity at the time, existence of surprise, whether the experience has been rehearsed (i.e., thought about or discussed), and presence of other people. The first 3 of these were called *canonical* characteristics by Brown and Kulik in their original study of the flashbulb memory because they were the most consistent elements of flashbulb memories.

The presumed role in memory played by some of these characteristics (rehearsal, for example) is wide ranging. Neisser (1982) suggested that recounting an experience may be a means for maintaining its freshness in memory as well as for endowing it with the various narrative characteristics just listed (e.g., description of location and presence of others).

Emotion, too, has been seen as playing a special role in memory. The special status accorded to emotion has been part of the common wisdom for two millennia (*To Gaius Herennius*, circa 10–20 BC./1954) and has been repeatedly confirmed by both clinical observation and laboratory experimentation. Although the importance of emotion as a factor in memory is widely recognized, the specific nature of its contribution is far from clear. On the one hand, emotion is sometimes seen to completely block the recall of a significant experience (Freud, 1926/1959) or to reduce the ability to recall certain features of an experience (Suengas & Johnson, 1988). On the other hand, it is sometimes seen to improve recall (Bower, 1981; Teasdale & Fogarty, 1979) or to contribute to the formation of a remarkably durable and vivid memory (Brown & Kulik, 1977). It is therefore difficult to anticipate the particular role of emotion in the establishment of ACEs. Certainly an analysis of ACE memories should include, in addition to consideration of the frequency with which affect is reported, a separate examination of positive and negative emotions and the pattern of linkage of these emotions to other ACE characteristics.

We expect that our data may contribute to a clarification of the phenomenon of reminiscence as discovered in individuals over age 50. Various experiments have found autobiographical memories to be most numerous in the immediate past, with a steady decline in frequency as the time line reaches into the past (Crovitz & Schiffman, 1974; Rubin, Wetzler, & Nebes, 1986).

The memories of people who are over 50 follow a different course; a *reminiscence surge* appears, defined as a peak in recall of experiences occurring between the ages of 10 and 30. Rubin et al. (1986) speculated that reminiscence (in this sense, a peak in memories in the past) is due to a process of "life review" in which subjects differentially sample memories from their earlier life (i.e., retrieve them unequally from different periods). They suggested that the basis of differential sampling will be better understood following a "detailed analysis of the memories that constitute the reminiscence component" (Rubin et al., 1986, p. 218); indeed, "reminiscence may require an expansion of standard cognitive models" (p. 220).

The broad concept of life review invoked by Rubin et al. (1986) is actually somewhat ambiguous in its implications for differential sampling. It could be taken to suggest that *all* earlier periods should be reviewed and sampled equally (rather than recent periods being sampled more heavily). We agree with Rubin et al. that life review could, however, involve systematic differential sampling. In particular, a life-review process may imply that life stages in which choices were made that are now seen to have affected the life course will be a special focus of memory sampling. If this is so, Eriksonian theory (see Erickson, Erickson, & Kivnick, 1986 for a summary) provides the basis for a specific prediction: Life narratives that focus on ACEs should exhibit a reminiscence effect, with a large proportion of memories concentrated during the period of identity formation. The application of Erikson et al.'s theory in this context illustrates the "expansion of standard cognitive models" envisioned by Rubin et al. Also, Erikson et al.'s theory of personality development leads us to expect the ACEs of the eminent psychologists we studied to be concentrated during late adolescence and young adulthood, rather than during the formative years of childhood and early adolescence, because it is during this later period that identity commitments are made.

In a recent, related argument, Fitzgerald (1988) suggested that the development of a relatively permanent "self-narrative" during the late adolescent and young adult years Erikson defined as *identity formative* might help account for reminiscence patterns. The creation of a stable account of one's own identity (what Fitzgerald considered a fixed self-narrative), and the making of enduring personal choices and commitments, are both features of Erikson's account of identity, and they are difficult to distinguish. However, if commitments and choices are experienced as entailing change, they might be expected to be reflected in the drawing of ACEs from times of life transition or change. Transitions are, moreover, generally viewed as points of affective and cognitive reorientation (Stewart, 1982; Stewart, Sokol, Healy, & Chester, 1986; Stewart, Sokol, Healy, Chester, & Weinstock-Savoy, 1982) and may therefore tend to be viewed as consequential in retrospect, although of course only some transitions are consequential. In fact, Pillemer et al. (1986) found that even for memories during the first year of college, a disproportionate number were associated with points of transition (the beginning of the academic year).

The goals of this study, then, were to (a) establish a method for coding memories for ACEs in personal documents; (b) assess the similarities of ACEs to other kinds of important autobiographical memories, such as flashbulb and vivid memories; (c) assess the role of affect in general, and positive and negative

emotions in particular, in ACE memories; and (d) assess the roles of life span developmental processes and other features of the life course, such as life transitions, in producing ACEs; these phenomena are expected to shed light on observed reminiscence patterns.

Method

Subjects

Autobiographical accounts of notable psychologists contained in three volumes of *A History of Psychology in Autobiography* (Boring & Lindzey, 1967; Lindzey, 1974, 1979) and *Models of Achievement: Reflections of Eminent Women in Psychology* (O'Connell & Russo, 1983) were the data analyzed in this study. These accounts were drawn from individuals in the same discipline (psychology), who were successful, whose careers spanned much of the 20th century, and who were old enough to have a relatively clear perspective on the consequential events of their lives. Moreover, all of the authors prepared their accounts for collections of autobiographies to be read by fellow professionals, and all were asked to cover roughly the same issues and types of points.

As we noted earlier, autobiographers were given considerable latitude in defining the scope of their accounts. For example, Boring and Lindzey (1967, p. vii) pointed out that "The important decisions in regard to the contents of your autobiography are yours." Moreover, autobiographers were encouraged to discuss "the historical details of your life," including their origins, education, and family, as well as ways in which their careers were influenced by "events, ideas, and persons in and out of the field of psychology" (p. vii). Boring and Lindzey (1967, p. vii) concluded

We should appreciate any discussion of your *feelings, motives, and aspirations* or of significant events that would increase the reader's understanding of you and your contributions to psychology. In brief, we are interested in your intellectual life history, but at the same time we feel that it should be illuminated by as much information about your personal background and inner motives as you are ready and able to divulge.

With few exceptions, autobiographers included discussion of their early lives and family backgrounds, emotional and psychological difficulties, and adult personal lives. The instructions to authors, then, seem quite unconstraining, and at the same time seem to point to a relatively broad consideration of the entire life course and the full range of personal and professional experience.

A few autobiographies in these volumes were not included in the study: three because the authors chose to write in the third person, making their memoirs difficult to code or to compare with the others; one because it was completed by another person posthumously; one because it was translated into English; and one because it was stated in the memoir that the author did not consider herself a psychologist. Ultimately, 31 male and 2 female autobiographers were included from the *History* volumes, and 16 female autobiographies were used from the *Models* volume, yielding a total of 49 autobiographies (of 31 men and 18 women). Subjects averaged about 72 years of age, with men slightly younger (about 70) than women (about 75); the youngest was 54, the oldest 86.

As noted earlier, the two sources were used to ensure inclusion of both men and women in the research from the beginning. However, any gender differences identified could result from differences in the sources. Therefore, these data could not be used as a basis for firm conclusions about gender; as discussed later, there were very few gender differences in the data.

Identifying ACEs

The first task was identification of memories for ACEs. To this end, the entire text of each psychologist's autobiography was read. A particular memory was coded as an ACE if it described an event, person, or set of circumstances in the individual's life that was remembered as having affected the unfolding of the life story in a personally significant way. Moreover, this consequentiality must have been evident through the writer's own words and could not be assumed. To illustrate, following are examples of ACEs in which the consequential component has been italicized. Some of these ACEs reflect episodic, one-moment-in-time events; others, more extended, nonepisodic events. This distinction is discussed later. (Because these quotations appear out of context, the full sense of the remarks may be missing.)

Since I had already been deeply impressed by subcultural differences between my Iowa grandparents on my father's side and my mother's California family, the emphasis on the cultural setting of behavior was absorbed into my bones; *it is impossible for me to write anything about development without attention to the subcultural context.* (L. B. Murphy, in O'Connell & Russo, 1983, p. 97)

I realized that my not raising a family *had given me freedom to make use of the opportunities that came my way*—even though I have been greatly involved in the lives of my large family and have often given personal matters priority over the professional ones. (E. Hanfmann, in O'Connell & Russo, 1983, p. 148)

I confessed my misgivings about my fitness to Professor Langfeld. In his laconic way he remarked, "But you know there are many branches of psychology." *I think this casual remark saved me.* (G. Allport, in Boring & Lindzey, 1967, p. 8)

The whole experience was a turning point. Never since that time have I been troubled by rebukes or professional slights directed at my maverick interests. (G. Allport, in Boring & Lindzey, 1967, p. 9)

To score as an ACE, the memory must also have established a causal connection between the event or experience and the consequence. This connection must have been explicitly made by the author, not merely suggested. Expressions frequently used to indicate causation include *because, effect, result, was due to, for that reason, therefore, and thus*. The following examples illustrate causality:

The undergraduate years *brought* a welter of new influences. (G. Allport, in Boring & Lindzey, 1967, p. 5)

An indelible impression *was made on me* then and later by Heraclitus, Socrates, Epicurus, and the Pythagoreans. . . . (G. Murphy, in Lindzey, 1974, p. 258)

The combination of my own experiences with children's behavior and thinking, along with getting acquainted with Erikson's and Murray's work, *led me* to my interest in studying child personality. . . . (L. B. Murphy, in O'Connell & Russo, 1983, p. 98)

By focusing only on ACEs in which an explicit, causal connection is made between a particular experience and its consequence, we no doubt excluded some events that were consequential. Given the paucity of existing research on ACEs, it seemed best to begin with a conservative operational definition, particularly one that would yield high levels of interrater agreement.

After identifying consequential events, the next step was to delineate all codable text relevant to the ACE. The beginning and ending of each ACE was determined by the divisions of the text provided by the autobiographer. In the simplest case, where the causal and consequential components of the ACE appeared within the same paragraph, that entire paragraph identified the ACE and provided the full memory text to be scored. In the following excerpt from Mead's autobiography (which was not included in our formal sample, because Mead was not

actually a psychologist), Woodworth's question to her and her reaction to it provides the cause and consequence of the ACE.

But I had never met Woodworth. He was on leave my senior year and I believe that if I had had him—how odd that idiom is—I would have remained in psychology. When I took his psychophysical methods [course], a subject in which I had little specific interest, we all sat transfixed as he made new discoveries with his back to the class, talking delightedly to himself. Later he was to ask me a question which set my feet on a path from which they have never strayed: "When does an Indian become an Indian?" Most of the behavioral science world still does not understand that a day-old baby is a perfect day-old sample of the culture within which it is born. I learned, too, from the way he carried the smallest detail along to find an error fifty pages later in the thesis, how to read a student's work. (M. Mead, in Lindzey, 1974, p. 311)

When the causal and consequential components were not contained within a single paragraph, often the case when the writer told the story of an influential experience in great detail, the ACE was extended to include the paragraphs containing the two components. If more than one ACE occurred within a single paragraph, the text within the paragraph was divided into shared and unshared components. Unshared text was specific to a single ACE and consisted of the statement of cause and consequence for that particular ACE; shared text consisted of everything else in the paragraph that was not excluded by this rule.

Interrater reliability was established in the following way: Two of us (W. Mackavey and A. Stewart) marked passages from a few autobiographies not included in the actual sample. Disagreements were discussed, and the guidelines described earlier were adopted to help us clarify exactly how ACE passages would be defined. Both raters then identified ACEs in 10 of the autobiographies actually used in the study, sampling from all sources. Passages were only counted as agreements if the raters agreed on (a) whether an ACE was contained in the material and (b) what material was to be included in the passage for coding. The agreement ($2 \times N$ agreements/ N scored by Mackavey + N scored by Stewart) on ACE identification was calculated for the 10 sources and was 89%. Mackavey then marked the remaining autobiographical texts for codable memories, indicating the beginning and ending of each ACE.

Content Analysis of ACEs

All ACEs were classified as to the type of memory they represented: episodic or nonepisodic. In an *episodic memory*, there is an explicit event or occurrence that occupies a particular, brief time frame, with a specific description of the event. Generally, episodic memories have a time span of a day or less. Memories that were either entirely episodic or that included an episodic element as a central feature of the memory were considered episodic.

In contrast, *nonepisodic* memories involved more general discussion of a life experience. These memories tend to extend over a time period longer than a day, and could range over several years, as when recalling one's marriage, early home life, college years, and so on. Two examples of the full text of ACEs drawn from Theodore M. Newcomb's autobiography illustrate the difference between episodic and nonepisodic memories. The first example includes a strong episodic component.

My formal transfer to Teachers College (which was willing to accept many of my academic credits from Union Seminary) was "decided upon" at a single moment, during my second year in New York. Every Tuesday and Thursday at 9:00 I left Goodwin Watson's class in "Psychology of Character" for a required class in "New Testament Exegesis" at the Seminary. At the end of the latter class, on a certain Tuesday, the professor, who had just devoted the entire hour to lecturing on the Greek word for "straightway" (found almost uniquely in the Gospel of St. Mark), announced

that he would need the full hour on Thursday, also, to lecture on that word. The contrast with the "Psychology of Character" (or was it the contrast with Watson?) was too much for me.

I "decided" at that moment that I would not complete the theological degree; I am not sure whether or not I ever returned to the New Testament class. At any rate Watson soon became, and remained, one of my two genuine mentors at Columbia. He was unfailingly provocative, learned, and helpful. I was, several years later, to have other kinds of close relationship with him. There was no one to whom I owed more, for some twenty years, and we still share interests and activities. (T. M. Newcomb, in Lindzey, 1974, p. 370)

In contrast, Newcomb also described a nonepisodic consequential memory:

My four predoctoral years were not as solitary as the preceding paragraphs might suggest. Almost every influence upon me, particularly while I lived in Union Seminary's dormitory, was relayed and reinforced, in the manner of social facilitation, by fellow students. Every exciting idea led to a sort of clan-gathering—especially ideas that stemmed from Messrs. Kilpatrick, Ward, Elliott, and Watson, as likely outside as in the classroom. Of the original five of us who were friends during undergraduate years and who influenced one another in coming to Union, I was to be the only one who moved into psychology. Fellow students whom I first met at or around the Seminary included Carl Rogers, Rensis Likert, and Ernest Hilgard. Here for the first time I experienced the exchange of intellectual stimulation closely related to what we were reading, or hearing in lectures, or hearing each other say in seminars. I am sure that these experiences had something to do with my present notions of what undergraduate education should be—and for me was not. (T. M. Newcomb, in Lindzey, 1974, p. 371)

Each ACE was also coded for various content categories drawn from the literatures on episodic memories and on reminiscence. Following are brief descriptions of these coding categories with examples. These descriptions are generally not sufficient for reliable coding; complete instructions are contained in a detailed coding manual that may be obtained from Mackavey.

Canonical Categories for Flashbulb Memories

Place. Coded yes if the memory includes mention of a specific place in which the author was physically located during the period recalled by the memory; otherwise no. C. H. Graham wrote, "As I sat in the dim ambient illumination of the dark room . . ." (in Lindzey, 1974, p. 104).

Own affect. Coded as positive or negative if the author explicitly mentioned some personal emotional experience as part of the ACE; otherwise coded no: "I did not weep: the tears simply flowed, as if unstoppered" (J. Bruner, in Lindzey, 1979, p. 79; coded negative).

Others' affect. Coded as positive or negative if the author explicitly attributed emotion to one or more other persons; otherwise coded no: "I dwell on this because it has probably made my attitude toward psychology dour and puritan, in a way that has sometimes offended younger colleagues" (D. Broadbent, in Lindzey, 1979, pp. 40–41; coded negative).

Ongoing activity. Coded yes if the ACE explicitly took place in the context of the author's ongoing activity; otherwise coded no. This category is codable only for memories with an episodic component: "Then one day I was walking home from the office with a friend whom I had previously recognized as a very tight fellow with a penny" (R. Sears, in Lindzey, 1979, p. 407).

Other Characteristics of Flashbulb or Vivid Memories

Rehearsal. Coded yes if the author indicated that the ACE had been thought about, discussed, or written about since its occurrence; other-

wise coded no. G. Allport said of his meeting with Freud, "I have told the story many times" (in Boring & Lindzey, 1967, p. 7).

Surprise. Coded yes if the author indicated that the consequence of the event or experience was unexpected; otherwise coded no. C. Rogers wrote, "When I wrote *Counseling and Psychotherapy* in 1942, neither my publisher or I could have foreseen that the miniscule field of counseling would suddenly expand at the end of the war into an enormous field of great public interest" (in Boring & Lindzey, 1967, p. 382).

Vividness. Coded yes if the memory is recalled with sufficient detail (e.g., sensory, spatial, or temporal detail, containing a verbatim quotation or described as vivid) that the memory has the character of a live experience; otherwise coded no: "I read the book so carefully that I found therein a crucial incidental passage, unnoticed by all reviewers, which almost got me out of bed shouting, 'Eureka!' " (D. Krech, in Lindzey, 1974, p. 226).

Other people. Coded yes if other people are mentioned; otherwise coded no. S. L. Pressey remembered: "Most broadly educative was a summer job in a department store; I made special deliveries all over town, including the red-light district, wrapped bundles, and chummed with fellow workers very different from people I had known before" (in Boring & Lindzey, 1967, p. 314).

Characteristics Associated With Reminiscence

Age. Each memory was scored for the approximate age of the subject at the time of the causal component of the ACE according to the following categories: early childhood (under 12 years old), early adolescence (13–17 years), college years (18–21 years), early adulthood (22–35 years), middle adulthood (36–50 years), and later adulthood (over 50 years). In some cases, age was uncodable. If an ACE spanned more than a single age period, the ACE period included all relevant age categories: "Thus far I have had eight more or less incapacitating depressions (1921, 1929, 1933, 1935, 1940, 1944, 1953, 1966)" (O. H. Mowrer, in Lindzey, 1974, p. 345).

Life transition. Coded yes if the causal component of the ACE occurred during a time of substantial life change (e.g., job change, travel, change in marital status, beginning college, becoming a parent, and significant health change); otherwise coded no: "Going off to college, then, was a critical point in my life—the shedding of an old skin for a new" (J. Bruner, in Lindzey, 1979, p. 79).

The procedure for establishing interrater reliability for the coding of categories was very similar to that described earlier for the identification of ACEs in the texts. After practice and clarification of category rules on a few autobiographies not included in the sample, and after attaining average interrater reliability above .85 (category agreement) for all categories with the other authors, all coding of ACE characteristics was carried out by Stewart.

Results

Types of ACEs

The autobiographies of the 49 subjects in this study yielded a total of 250 ACEs ($M = 5.1$, range = 1–9). Most ACEs (173 of 250) were of the nonepisodic type. Forty memories (16%) were strictly episodic, and an additional 37 (15%) had some episodic component; thus, 31% of the memories were treated as episodic, whereas the remaining 69% only described events occurring over relatively long or diffuse periods.

Content Analysis of ACEs

All memories were coded for all content categories except for the category ongoing activity, which applied only to episodic

memories. In Table 1, the characteristics of all ACEs, only episodic ACEs, and only nonepisodic ACEs are listed. The table shows the percentages of ACEs containing each of the scoring categories, together with the results of a chi-square analysis of the significance of the difference in these percentages between episodic and nonepisodic ACEs. First, the most characteristic feature of all ACEs was the presence of other people (92% included them). Also relatively common were two of the canonical categories for flashbulb memories: place (54%) and own affect (30%), although others' affect was rather rare (9%). It was quite common for ACEs to be described as occurring at life transitions (42%). Rehearsal and surprise were rare.

Episodic and nonepisodic memories differed in several ways. Episodic memories were even more likely than nonepisodic ones to take place during a life transition, to be described in vivid terms, or to include references to place and own affect (two canonical categories for flashbulb memories). In fact, 80% of the episodic memories contained at least one of the canonical categories for flashbulb memories, whereas only 60% of the nonepisodic memories did, $\chi^2(1, N = 250) = 9.5, p < .01$. In addition, episodic memories were more likely to involve surprise, $\chi^2(1, N = 250) = 6.08, p < .01$.

Distribution of ACEs Across the Life Span

Table 2 displays the percentage of ACEs associated with each of the age categories. Because some memories spanned more than one period, the total sometimes exceeds 100. Both episodic and nonepisodic memories occurred with maximum frequency during the years of early adulthood (22–35); about half of all ACEs were drawn from this period. Because the age categories do not contain equal numbers of years, the number of ACE memories per year within each of the categories is also shown for descriptive purposes. On this basis, the most fertile period for both memory types was the college years. Thus, about 80% of all ACEs were identified with ages 18–35. Interestingly, relatively few ACEs were drawn from either the suppos-

edly formative early childhood years or the most recent years of the autobiographers' lives (after age 50).

Episodic and nonepisodic memories differed in the age from which they were drawn in two ways. The Yates-corrected chi-square analysis of the percentage of episodic and nonepisodic memories occurring within each of the age categories revealed that nonepisodic memories were more frequently based on early adolescent (13–17) and middle adulthood (36–50) experiences than were episodic memories. This is almost certainly due to the tendency for nonepisodic memories to span more than one age period and thus to extend beyond the two most frequently mentioned periods.

The Role of Affect

The role of affect as a key marker for consequentiality was explored by assessing the relationship between reports of affect and the other content categories. Yates-corrected chi-squares were calculated on reports of one's own affect with reports of others' affect in terms of the overall presence of affect as well as the presence of positive or negative affect. As shown in Table 3, if the autobiographers we studied mentioned their own affect, they were more likely to include a description of another person's affect. It was unusual for them to mention another person's affect while remaining silent about their own. The significance of the effect was evaluated by means of a Yates-corrected chi-square.

As shown in Table 4, overall mention of own affect was also associated with several other memory characteristics, although the pattern was different for the recall of positive and negative affect. According to the Yates-corrected chi-square analysis, memories containing references to vivid experiences, rehearsal, life transitions, and place were more likely to contain a reference to affect than were those that did not. They were also *less* likely to be drawn from the period of early adulthood.

However, these results were different for positive and negative affect memories. Only positive affect memories were more

Table 1
Characteristics of Memories for Autobiographically Consequential Experiences (ACEs)

Memory category	Percentage of:			$\chi^2(1)$
	All ACEs (<i>N</i> = 250)	Episodic ACEs (<i>n</i> = 77)	Nonepisodic ACEs (<i>n</i> = 173)	
Canonical (flashbulb)				
Place	54	66	49	6.01*
Own affect	30	44	23	10.33*
Others' affect	9	10	8	0.12
Ongoing activity	—	15	—	—
Other flashbulb and vivid				
Rehearsal	9	13	7	1.74
Surprise	10	18	7	6.08**
Vividness	18	42	7	41.69***
Other people	92	94	91	0.11
Reminiscence related				
Life transition	42	60	34	13.34***

Note. The Yates-corrected chi-square was used to test significance of differences. Dashes indicate data were not applicable.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 2
Age Distribution of Memories for Autobiographically Consequential Experiences (ACEs)

Category	Age (years)	n	Episodic ACEs		Nonepisodic ACEs		$\chi^2(1)$
			%	% per year	%	% per year	
Childhood	<12	28	6.49	0.54	13.77	1.77	2.08
Adolescence	13-17	32	3.90	0.78	17.37	3.47	7.25*
College	18-21	70	33.77	8.44	26.35	6.59	1.08
Early adult	22-35	124	49.35	3.53	51.50	3.68	0.03
Middle adult	36-50	46	9.09	0.61	23.35	3.43	6.11*
Later adult	>50	20	5.19	0.24	9.59	0.44	0.83

Note. Yates-corrected chi-squares were used to test significance of differences.

* $p < .01$.

vivid and included reference to place. They were also likely to be drawn from the college years and not from early adulthood. In contrast, only negative affect memories tended to be more often rehearsed, and they took place at transitions.

Gender

There were very few gender differences in ACE features. Men produced more ACEs ($M = 5.90$ for men, $M = 3.72$ for women), $t(47) = 2.61$, $p < .05$, probably because men wrote longer autobiographies (mean number of pages was 30.48 for men and 14.33 for women), $t(47) = 5.98$, $p < .001$. The only other differences, though, were that men were more likely to report rehearsal of their ACEs (mean proportion of own memories rehearsed was 17 for men and 6 for women), $t(47) = 2.66$, $p < .05$, and to mention ongoing activity (mean proportion making such references was 5.8 for men and 0.6 for women), $t(47) = 2.35$, $p < .05$. There were no differences in any other categories.

Discussion

This study aimed to illustrate, encourage, and broaden the use of literary or historical documents as a data source for the quantitative analysis of psychologically significant problems (Healy & Stewart, in press). We examined the professional autobiographies of several distinguished psychologists in order to understand how an important naturally occurring experience, the ACE, is represented in memory. We found that ACEs tended

to be remembered as occurring over extended periods, not in particular episodes. In this respect, ACEs stand apart from the experiences that make up the vast majority of the autobiographical memory literature. Although it is not yet clear whether ACEs focusing on areas of life other than career would be similarly divided, we suspect that temporally extended, or nonepisodic, memories would continue to dominate. There is other evidence that when subjects are queried with open-ended questions about their past, nonepisodic memories are reported more frequently (Pillemer et al., 1986). The essentially non-directive nature of the instructions given to our subjects, with respect to our categories, probably yielded underestimates of the recoverability of the memory details listed in Table 1. More focused questions about affect, or location in space and time, might have resulted in fuller descriptions.

Similarity of ACEs to Flashbulb or Vivid Memories

ACEs often included some of the defining characteristics of flashbulb memories (notably references to place and own affect). However, this was much more true of episodic than of nonepisodic memories, and these were the rarer ACEs. Similarly, episodic ACEs were often vivid, but nonepisodic ACEs were not. It is possible that this difference is due to the inherent difficulty of forming a perceptual abstraction, or image, of a temporally extended or generic experience (Brewer, 1986).

The one central characteristic of nearly all ACEs seemed to

Table 3
Associations Between Reports of Own and Others' Affect in Autobiographically Consequential Experiences

Other	Own total affect			Own positive affect			Own negative affect		
	Present (%)	Absent (%)	$\chi^2(1)$	Present (%)	Absent (%)	$\chi^2(1)$	Present (%)	Absent (%)	$\chi^2(1)$
Total affect	20.3	4.0	15.26**	20.8	5.9	8.94**	19.4	7.3	3.53
Positive affect	10.8	2.3	6.55*	14.6	2.5	9.93**	6.5	4.6	0.00
Negative affect	12.2	2.3	8.43**	8.3	4.5	0.53	16.1	3.7	6.23*

Note. Yates-corrected chi-squares were used to test significance of associations. $N = 250$ autobiographically consequential experiences.

* $p < .05$. ** $p < .01$.

Table 4

Associations Between Affect Reports and Other Characteristics of Memories for Autobiographically Consequential Experiences

Memory category	Own affect total			Own positive affect			Own negative affect		
	Present (%)	Absent (%)	$\chi^2(1)$	Present (%)	Absent (%)	$\chi^2(1)$	Present (%)	Absent (%)	$\chi^2(1)$
Canonical (flashbulb)									
Place	67.6	48.3	7.03**	68.8	50.5	4.49*	67.6	52.1	2.10
Ongoing activity	9.1	22.2	0.24	14.3	15.4	0.66	10.0	16.7	0.53
Other flashbulb and vivid									
Rehearsal	17.6	5.1	8.58*	14.6	7.4	1.66	22.6	6.8	6.53*
Vividness	29.7	12.5	9.51**	29.2	14.9	4.54*	29.0	16.0	2.35
Surprise	14.9	8.5	1.62	16.7	8.9	1.74	12.9	10.0	0.03
Other people	94.6	90.9	0.53	97.9	90.6	1.92	90.3	92.2	0.00
Reminiscence related									
Life transition	56.8	35.8	8.56**	45.8	41.1	0.19	74.2	37.4	13.59***
Age									
Childhood	12.3	11.1	0.00	10.6	11.7	0.00	12.9	11.3	0.00
Adolescence	12.3	13.5	0.00	12.8	13.2	0.03	9.7	13.6	0.10
College	37.0	25.1	2.95	48.9	23.9	10.47**	25.8	29.1	0.03
Early adult	38.4	56.1	5.78*	31.9	55.3	7.41**	45.2	51.6	0.23
Middle adult	17.8	19.3	0.01	17.0	19.3	0.02	16.1	19.2	0.03
Later adult	4.1	9.9	1.60	4.3	9.1	0.64	3.2	8.9	0.52

Note. Yates-corrected chi-squares were used to test significance of associations. $N = 250$ autobiographically consequential experiences.

* $p < .05$. ** $p < .01$. *** $p < .001$.

be references to other people. The prominence of other people in ACEs is consistent with the figures reported by Pillemer et al. (1986) for the memories of college students, and those reported by Hall and Van de Castle (1966) for inclusion of other people in dream reports. References to location in space, and to life transitions, were fairly frequent in both types of ACEs, but much more frequent in episodic than in nonepisodic ones. The common mention of place in episodic memories may reflect narrative convention (Neisser, 1982) and the fact that a single place may encompass the entire experience. Finally, for the theoretical reasons discussed in the following section on reminiscence, life transitions provided increased opportunities for the occurrence of ACEs, but it is not entirely clear why so many more episodic memories were associated with life transitions than nonepisodic ones. Future research might aim at clarifying the sources of the many differences between episodic and nonepisodic ACEs (five out of nine categories, of which four occurred quite frequently overall).

Gender and ACEs

There were few gender differences in ACEs within this extremely selective sample. The differences that were observed might be taken to suggest that men's autobiographical accounts reflected a greater self-focus (greater length, more ACEs, more rehearsal, and more mention of own activities at the time) or sense of the importance of events in one's own life. This self-focus may be generally encouraged in men—particularly achieving men—but is apparently less encouraged even in achieving women. On the other hand, this interpretation should be viewed as quite tentative because differences in the two sources (for most women vs. all men) may have produced the apparent gender differences.

Reminiscence

The greatest concentration of memories for our subjects did not occur during the recent time of their lives, but rather some 50 or so years earlier. In this respect, the form of these data is remarkably different from the monotonically declining remembering functions regularly reported under both cued- and free-recall conditions (Crovitz & Schiffman, 1974; Rubin et al., 1986). Rubin et al. (1986, p. 208) used the term *reminiscence* to refer to the "increase in early memories above what would be expected by a monotonically decreasing retention function." Following a detailed analysis of cued-recall data from several laboratories, they concluded that reminiscence is absent from the data of subjects younger than 30. However, "by age 50, there is reminiscence which is as pronounced as it is with 60- and 70-year-old subjects" (Rubin et al., 1986, p. 208). In their secondary analysis, there was a surge in the frequency of memories from the approximate ages of 10 to 30 years.

Our data suggest a broader conception of reminiscence. The surge in early memories was not superimposed upon a monotonically declining retention function. The ACE memories of our subjects became relatively more frequent, not less frequent, as the time line stretched into the past; they were most numerous from the ages of 22 to 35; then, at even younger ages, they declined in frequency once more. The "reminiscence bump" in the context of this distribution is very striking. Rubin et al. (1986) considered the "differential sampling hypothesis" the best explanation of their reminiscence findings. They suggested that early life experiences in an older population would be more heavily sampled because these subjects are apt to be engaged in a process of life review. Fitzgerald (1988) offered an alternative account of the reminiscence bump. He suggested that the form of the curve, especially the bump, reflects the underlying age distribution of highly accessible vivid memo-

ries. The age distribution of these salient autobiographical memories was clustered around the adolescent and early adult years because that is when individuals construct their self-narrative, which is "a contextualized view of the self composed of stories and tales rather than traits and social roles" (Fitzgerald, 1988, p. 261). Later in life, during adulthood, the self-narrative changes less and incorporates fewer new items, and the age distribution of vivid memories stabilizes at a lower level.

The age distribution of our ACE-based memories, with its lack of a recency component, resembles the shape of Fitzgerald's (1988) curve. That, of course, suggests the possibility that a similar mechanism might underlie our two distributions. Most of our ACE-based memories, however, differ in one essential respect from those of Fitzgerald's subjects: Most of our ACE-based memories were not vivid memories. Indeed, as already mentioned, ACE memories are based on experiences that extend over long periods and as such are not easily abstracted and coded in perceptual terms (Brewer, 1986). Thus, Fitzgerald's explanation, which is based on an assumption of vividness in the memories, cannot account for our data. For this reason, we prefer an explanation for our results based on the differential sampling hypothesis. The central question, then, becomes why the most extensive sampling occurred from the college and early adult periods of their lives.

Erikson's theory of personality development, as indicated earlier, seems to be able to account for both sets of findings. Thus, Erikson et al. (1986) discussed a process quite similar to that of life review and suggested that

those nearing the end of the life cycle find themselves struggling to accept the inalterability of the past and the unknowability of the future, to acknowledge possible mistakes and omissions, and to balance consequent despair with the sense of overall integrity that is essential to carrying on. (p. 56)

They pointed out that "An important component of this process is the acknowledging and accepting of past choices, made along life's course" (p. 70). Accordingly, individuals of old age will be expected to focus on the period that was most formative of the adult life course and that involved the largest number of unalterable choices and decisions.

From Erikson et al. (1986), we know that the developmental period most associated with the formation of an occupational identity is the period of transition from adolescence to adulthood. For our subjects (as perhaps for many career-involved adults), this would be the time when they were in college and graduate school and making decisions about their first professional positions. Thus, this period was both a formative one and one filled with life transitions that might provide the stimulus for ACE memories.

Erikson's theory, in our view, provides the best and most coherent framework for understanding our data. There are, of course, alternative approaches, although none seem as complete or as satisfying as Erikson's. For example, psychodynamic theorists, with their emphasis on the personality-formative influence of early family life, might predict that the greatest contribution to the memory pool would come from experiences occurring during the formative years of childhood and adolescence. Such a prediction, of course, would not have been confirmed. Similarly, a social psychological emphasis on the demand characteristics of the instructions to our subjects might yield a prediction that career-related ACEs would emerge most often after childhood and before late adulthood; it would not,

though, account for the fact that our subjects attributed greater consequentiality to the experiences of their college and early adulthood years rather than either adolescence or the middle adulthood years.

In a somewhat different vein, the instructions given to our subjects could be construed as encouraging the report of episodic experiences (e.g., "We should appreciate any discussion . . . of significant events . . ."). Despite this, nonepisodic memories emerge as more frequent at every age level. We interpret this to mean that ACEs are often experiences without precise moments of beginning and without sharp endings. Indeed, they may continue for years. Whether this is in fact true for ACEs from other areas of life as well is an unanswered question. If the pattern of our data in this respect is confirmed in research with other samples, then a broadening of current autobiographical research with its heavy emphasis on episodic experiences would seem appropriate. It will, however, be necessary to assess the extent to which our findings are a function of sample characteristics, instructions given, and the age or life stage of the autobiographers. Collection of new, or examination of existing, autobiographies from a wide range of individuals in terms of age, career, and social position, and written in accordance with a variety of instructions, would be a useful next step.

The Role of Affect

Finally, affect played a complex role in the ACE memories of our subjects. The association between emotions and other ACE characteristics differed for memories having positive or negative affect. Only negative affect, for example, is associated with rehearsal and life transitions. Moreover, although references to affect were relatively rare, they were more likely to be associated with many other details. This may be due to the professional and public nature of the documents involved. Only additional research, with different types of autobiographies, could clarify this.

However, affectively charged ACEs were more likely to include several other details; thus, affect recall does not seem incompatible with sensory recall. This naturalistic outcome stands in contrast to the laboratory-based findings of Suengas and Johnson (1988). They reported that when subjects were *instructed* to focus on "apperceptive" (i.e., positive and negative feelings) features of their experiences in the lab, they tended not to recall perceptual features of those experiences. They reported that

thinking or talking about apperceptive aspects of events . . . may reduce accessibility of sensory and contextual characteristics of memories. . . . Emotion could produce apparent "repression" because (relative to an unemotional event) perceptual features of the event do not receive the attention they otherwise would have received. (Suengas & Johnson, 1988, p. 388)

This reasoning is, of course, difficult to square with our findings for ACEs, or with Brown and Kulik's (1977) suggestion that with flashbulb memories it is partly affect that cues the subject to "take a visual picture" of the situation he or she is in. At least in the case of a life experience of central importance, and an experimentally unconstrained opportunity to rehearse the experience, the presence of an emotional component may contribute to the vividness of the memory (Rubin & Kozin, 1984). In this respect, our finding of a significant association between the vividness of a memory and its emotional content is consistent

with the spirit of the remarks made by the anonymous Roman orator who observed:

When we see in everyday life things that are petty, ordinary, and banal, we generally fail to remember them. . . . But if we see or hear something exceptionally base, dishonourable, extraordinary, great, unbelievable, or laughable, that we are likely to remember a long time. (*To Gaius Herennius*, circa 10–20 B.C./1954)

This tendency will only be enhanced if, with hindsight, we view that same experience as consequential for the later course of our lives.

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